14-10-2026: Linux Commands

```
Anudeep:~# mkdir -p ~/linux_advanced_lab/logs
Anudeep:~# cd ~/linux_advanced_lab
Anudeep:~/linux_advanced_lab# cat > logs/application.log <<'EOF'
> 2026-10-14 INFO Application started
> 2026-10-14 ERROR Database connection failed
> 2026-10-14 WARNING High memory usage
> 2026-10-14 ERROR API request failed
> 2026-10-14 INFO Application recovered
> EOF
Anudeep:~/linux_advanced_lab# cat > users.txt <<'EOF'
> anudeep:1001:Developer
> rahul:1002:Tester
> priya:1003:Admin
> service:500:System
> EOF
Anudeep:~/linux_advanced_lab# cat > application.sh <<'EOF'
> #!/bin/bash
> echo "Application deployment started"
> echo "Checking server configuration"
> echo "Deployment completed"
> EOF
Anudeep:~/linux_advanced_lab# cat > app.conf <<'EOF'
> APP_NAME=Marketplace
> APP_PORT=8080
> APP_ENV=development
> EOF
Anudeep:~/linux_advanced_lab# echo "Linux lab created successfully"
Linux lab created successfully
Anudeep:~/linux_advanced_lab# find ~/linux_advanced_lab -maxdepth 2 -type f
/root/linux_advanced_lab/logs/application.log
```

```
/root/linux_advanced_lab/app.conf
/root/linux_advanced_lab/application.sh
/root/linux_advanced_lab/users.txt
```

1. find — Search for files by name and modification time

   Description :Searches the lab directory for log files modified within approximately the last seven 24-hour periods.

```
Anudeep:~/linux_advanced_lab# find . -type f -name "*.log" -mtime -7
./logs/application.log
```

2. awk — Extract specific fields from a file

   Description: Reads users.txt, uses : as the field separator, and prints entries whose second field is at least 1000.

```
Anudeep:~/linux_advanced_lab# awk -F: '$2 >= 1000 {print "User:", $1, "ID:", $2}' users.txt
User: anudeep ID: 1001
User: rahul ID: 1002
User: priya ID: 1003
```

3. sed — Replace text in file output
   Description: Replaces every occurrence of ERROR with CRITICAL in the displayed output. The original file remains unchanged.

```
Anudeep:~/linux_advanced_lab# sed 's/ERROR/CRITICAL/g' logs/application.log
2026-10-14 INFO Application started
2026-10-14 CRITICAL Database connection failed
2026-10-14 WARNING High memory usage
2026-10-14 CRITICAL API request failed
2026-10-14 INFO Application recovered
```

4. lsof — Identify processes using a file
   First, open the log file in a background process that keeps it open:

```
Anudeep:~/linux_advanced_lab# tail -f logs/application.log &
Anudeep:~/linux_advanced_lab# 2026-10-14 INFO Application started
2026-10-14 ERROR Database connection failed
2026-10-14 WARNING High memory usage
2026-10-14 ERROR API request failed
```

   Then run:

```
lsof logs/application.log
1       /init   0       /dev/null
1       /init   1       /dev/null
1       /init   2       /dev/kmsg
1       /init   3       /dev/hvc1
1       /init   4       mnt:[4026532217]
1       /init   5       socket:[1104]
1       /init   7       socket:[7176]
1       /init   8       socket:[7177]
1       /init   100     socket:[4200]
5       /init   0       /dev/null
5       /init   1       /dev/null
5       /init   2       /dev/kmsg
5       /init   3       /dev/kmsg
5       /init   4       /dev/hvc1
5       /init   5       /
5       /init   6       socket:[1105]
5       /init   7       socket:[1106]
5       /init   8       anon_inode:[eventpoll]
11      /init   0       /dev/null
11      /init   1       /dev/null
11      /init   2       /dev/kmsg
11      /init   3       /dev/hvc1
11      /init   4       mnt:[4026532217]
11      /init   5       socket:[1104]
11      /init   6       socket:[7187]
11      /init   7       socket:[7176]
11      /init   8       socket:[7177]
11      /init   9       socket:[639]
12      /init   0       /dev/null
12      /init   1       /dev/null
12      /init   2       /dev/kmsg
12      /init   3       /dev/hvc1
12      /init   4       mnt:[4026532217]
12      /init   5       socket:[1104]
12      /init   6       socket:[7187]
```

   Description: Displays processes that currently have the log file open, including process IDs and access modes.

```
13      /bin/busybox    0       /dev/pts/0
13      /bin/busybox    1       /dev/pts/0
13      /bin/busybox    2       /dev/pts/0
13      /bin/busybox    10      /dev/tty
25      /init   0       /dev/null
25      /init   1       /dev/null
25      /init   2       /dev/kmsg
25      /init   3       /dev/hvc1
25      /init   4       mnt:[4026532217]
25      /init   5       socket:[1104]
25      /init   6       socket:[7211]
25      /init   7       socket:[7176]
25      /init   8       socket:[7177]
25      /init   9       socket:[1204]
26      /init   0       /dev/null
26      /init   1       /dev/null
26      /init   2       /dev/kmsg
26      /init   3       /dev/hvc1
26      /init   4       mnt:[4026532217]
26      /init   5       socket:[1104]
26      /init   6       socket:[7211]
26      /init   7       socket:[7176]
26      /init   8       socket:[7177]
26      /init   9       socket:[1204]
26      /init   10      socket:[4349]
26      /init   11      socket:[4344]
26      /init   12      socket:[4345]
26      /init   13      socket:[4346]
26      /init   14      socket:[4347]
26      /init   15      socket:[4348]
26      /init   16      /dev/ptmx
26      /init   17      anon_inode:[signalfd]
26      /init   18      /dev/ptmx
27      /bin/busybox    0       /dev/pts/1
27      /bin/busybox    1       /dev/pts/1
27      /bin/busybox    2       /dev/pts/1
27      /bin/busybox    10      /dev/tty
38      /bin/busybox    0       /dev/pts/1
38      /bin/busybox    1       /dev/pts/1
38      /bin/busybox    2       /dev/pts/1
38      /bin/busybox    3       /root/linux_advanced_lab/logs/application.log
```

Creating a directory as linux_day2_lab

```
Anudeep:~# mkdir -p ~/linux_day2_lab
Anudeep:~# cd ~/linux_day2_lab
Anudeep:~/linux_day2_lab# cat > server_metrics.txt <<'EOF'
> CPU 78
> Memory 64
> Disk 91
> CPU 45
> Memory 72
> Disk 55
> EOF
Anudeep:~/linux_day2_lab# cat > deployment.env <<'EOF'
> APP_NAME=Marketplace
> APP_PORT=8080
> APP_ENV=development
> EOF
Anudeep:~/linux_day2_lab# cat > services.txt <<'EOF'
> nginx running
> docker stopped
> ssh running
> database running
> redis stopped
> EOF
Anudeep:~/linux_day2_lab# mkdir reports
eport" > reports/report.txt
echo "Backup report"Anudeep:~/linux_day2_lab# echo "Deployment report" > reports/report.txt
Anudeep:~/linux_day2_lab# echo "Backup report" > reports/backup.txt
```

5. sort — Sort data numerically or alphabetically

   Description: Sorts the server metrics by the second column in descending numerical order.

- -k2,2 — sorts using the second field.
- -n — compares values numerically.
- -r — reverses the order.

```
Anudeep:~/linux_day2_lab# sort -k2,2nr server_metrics.txt
Disk 91
CPU 78
Memory 72
Memory 64
Disk 55
CPU 45
```

6.uniq — Identify repeated lines

First, create a file containing duplicate events:

```
Anudeep:~/linux_day2_lab# printf 'LOGIN\nERROR\nLOGIN\nERROR\nLOGIN\nSUCCESS\n' > events.txt
Anudeep:~/linux_day2_lab# sort events.txt | uniq -c
      2 ERROR
      3 LOGIN
      1 SUCCESS
Anudeep:~/linux_day2_lab# cut -d= -f1 deployment.env
APP_NAME
APP_PORT
APP_ENV
```

Description: Counts consecutive duplicate lines after sorting the input.

7.cut — Extract selected fields

Description: Extracts the configuration variable names from a file.

- -d= — uses the equals sign as the delimiter.
- -f1 — selects the first field.

```
Anudeep:~/linux_day2_lab# cut -d= -f1 deployment.env
APP_NAME
APP_PORT
APP_ENV
```

8.xargs — Pass input to another command

Description: Reads input and supplies each item as an argument to another command.

- -I {} — defines {} as the replacement placeholder.
- echo — displays the generated message.

```
Anudeep:~/linux_day2_lab# printf 'report.txt\nbackup.txt\n' | xargs -I {} echo "Processing file: {}"
Processing file: report.txt
Processing file: backup.txt
Anudeep:~/linux_day2_lab# cat > deployment_v2.env <<'EOF'
> APP_NAME=Marketplace
> APP_PORT=9090
> APP_ENV=production
> EOF
```

9. diff — Compare two files

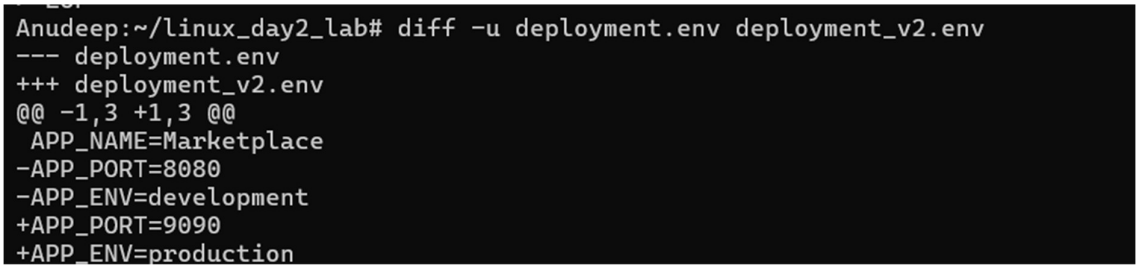
Description: Displays the differences between two files in unified format.

- -u — shows changes with surrounding context.
- - lines — indicate content removed from the first file.
- + lines — indicate content added in the second file.

Create a second configuration file:



Compare the configurations:



10. ln — Create a symbolic link

Description: Reports the total disk usage of the reports directory.

- -s — shows a summary instead of individual entries.
- -h — displays human-readable sizes.

```
Anudeep:~/linux_day2_lab# du -sh ~/linux_day2_lab/reports
3.0K    /root/linux_day2_lab/reports
Anudeep:~/linux_day2_lab# df -h ~/linux_day2_lab
Filesystem                Size      Used Available Use% Mounted on
/dev/sdd                128.9M     65.2M     53.7M  55% /
```

free — Monitor memory utilization

Description: Displays system RAM and swap memory statistics.

- total — total memory.
- used — memory currently used, as reported by the tool.
- free — completely unused memory.
- available — estimated memory available for new applications without swapping.

```
Anudeep:~/linux_day2_lab# free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6G      340.6M        7.2G        3.7M       89.3M        7.1G
Swap:          2.0G           0        2.0G
Anudeep:~/linux_day2_lab# uptime
 09:49:41 up  2:55,  0 users,  load average: 0.07, 0.01, 0.00
```